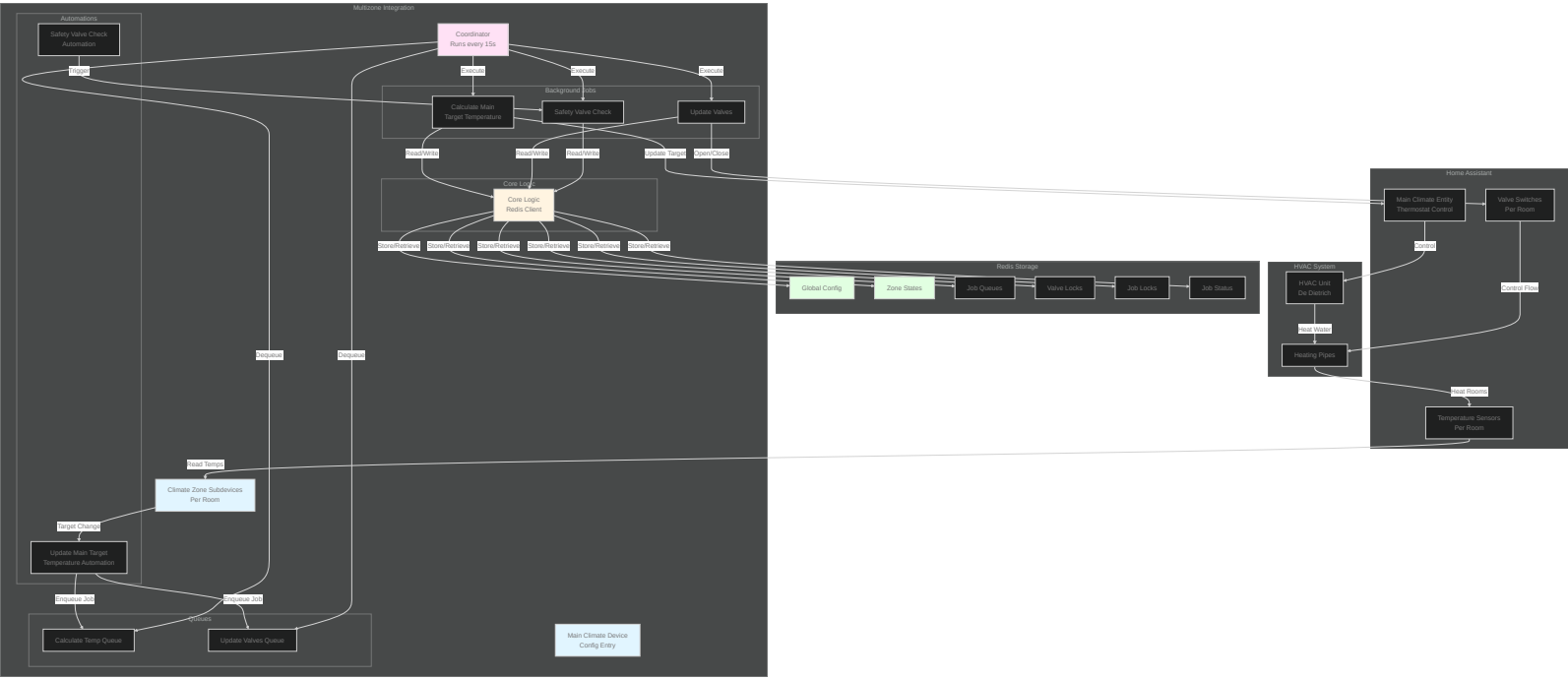




Start: Calculate Main Target



Yes

Filter: Get Active Zones
state != OFF



Yes

Filter: Exclude Overheated
Zones
satisfaction != overheated

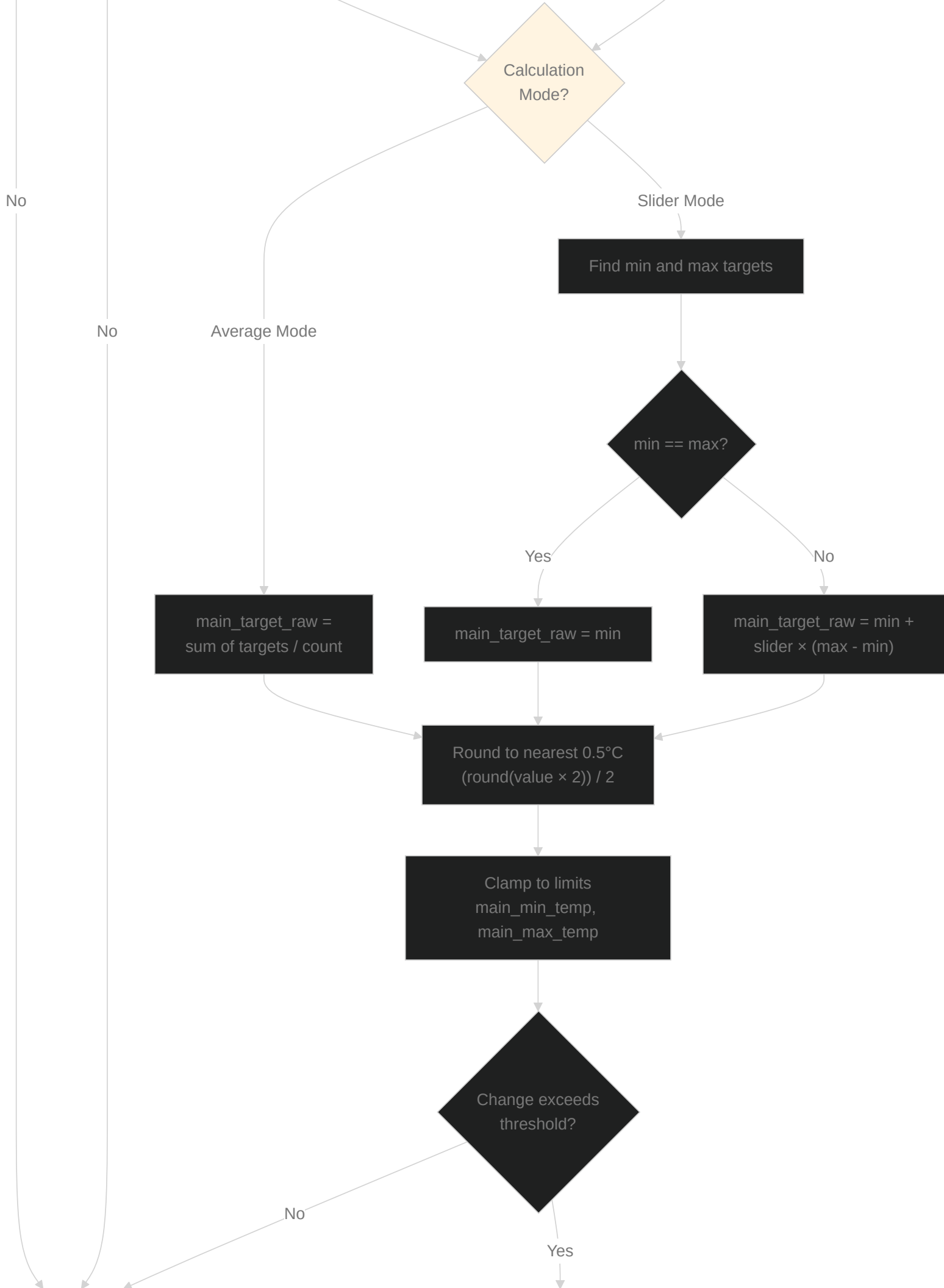


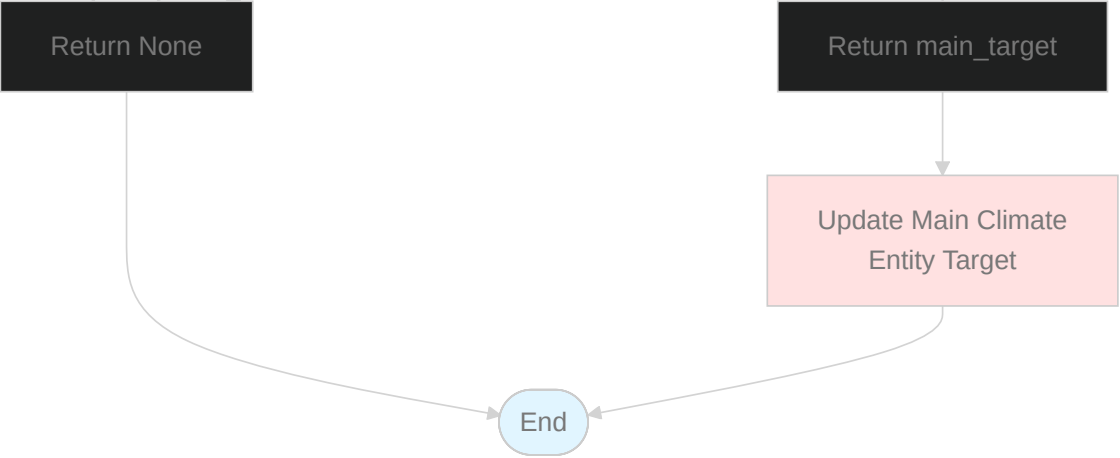
Yes

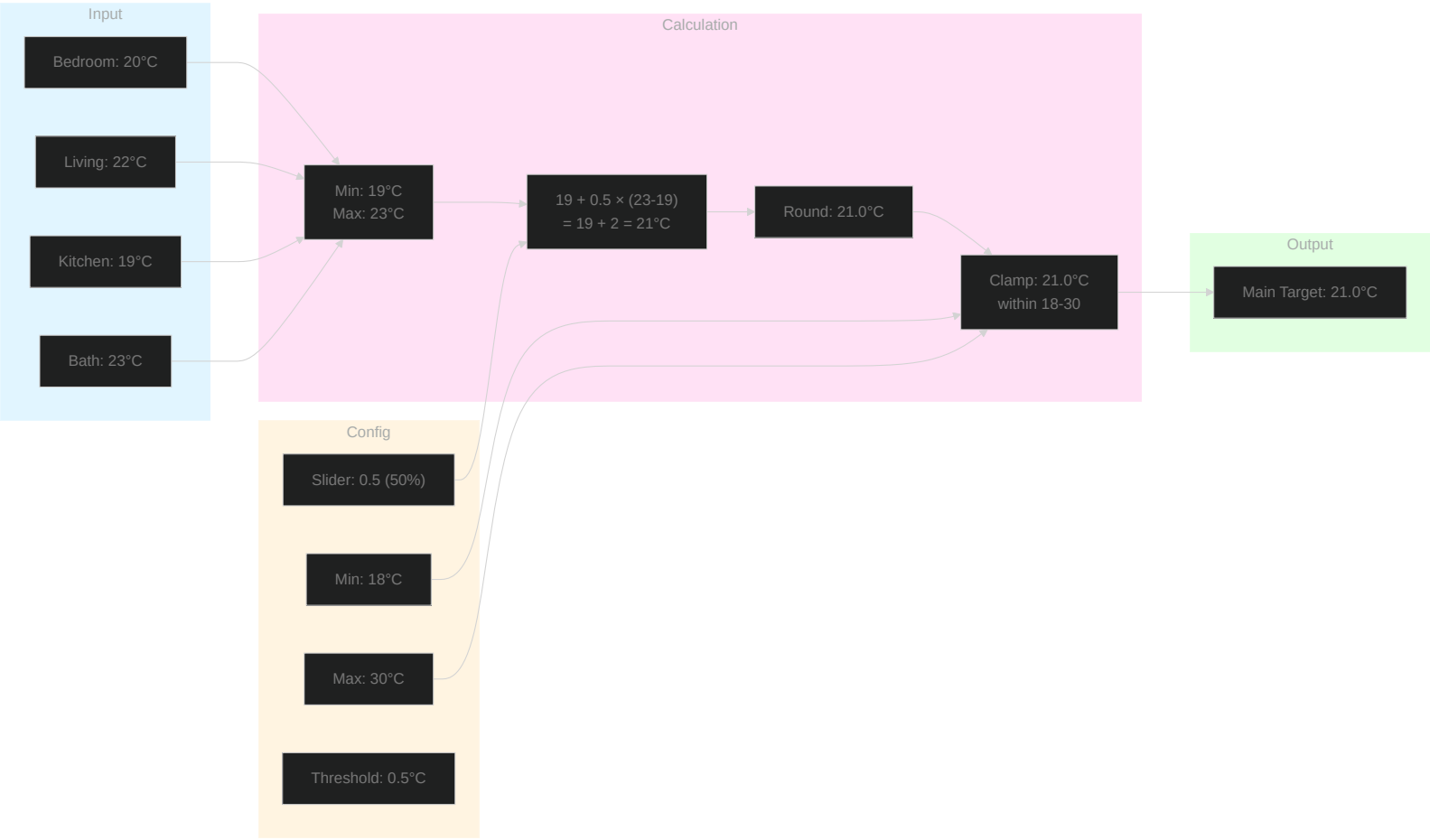
Use Non-Overheated Zone
Targets

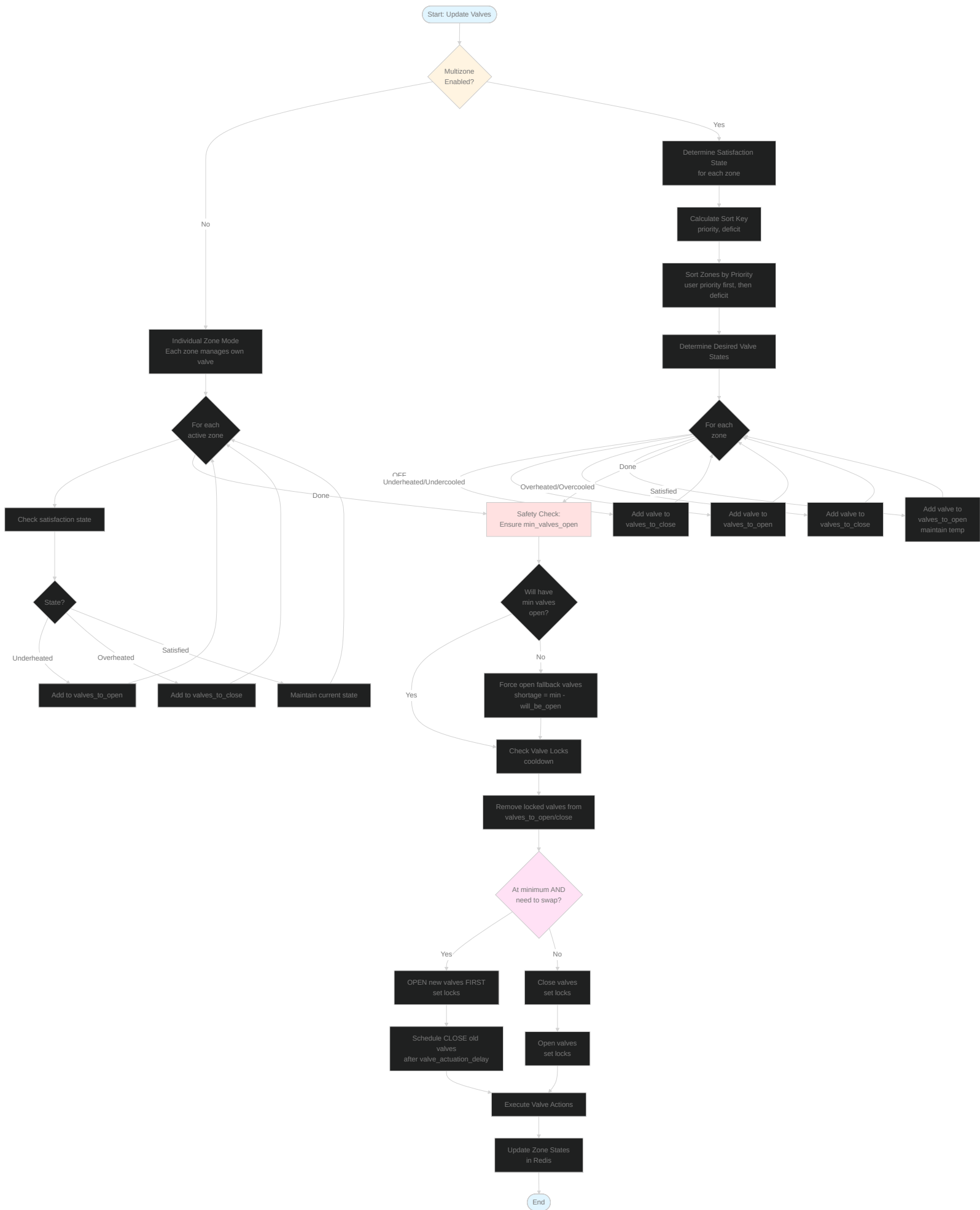
No

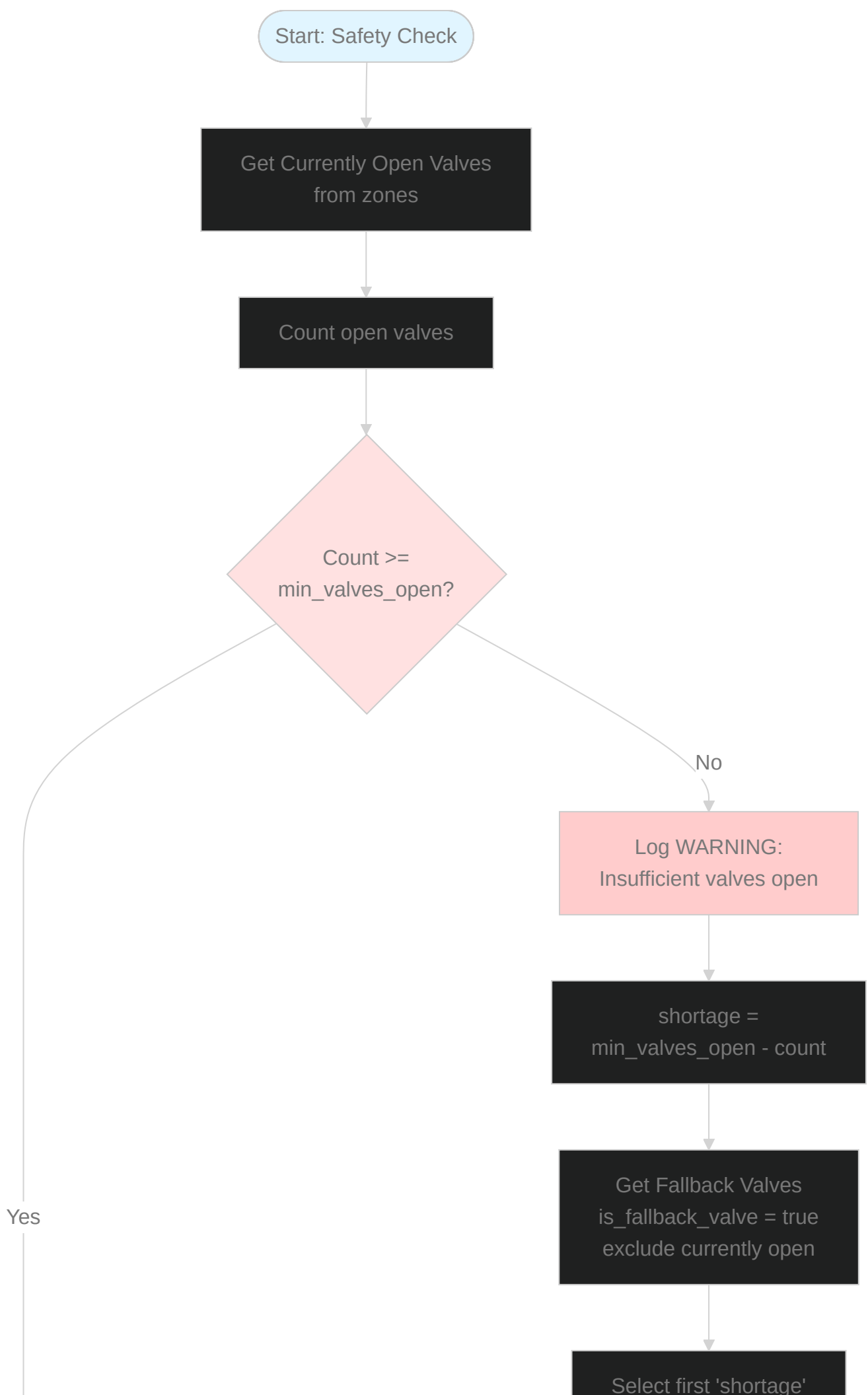
Use All Active Zone Targets
Fallback

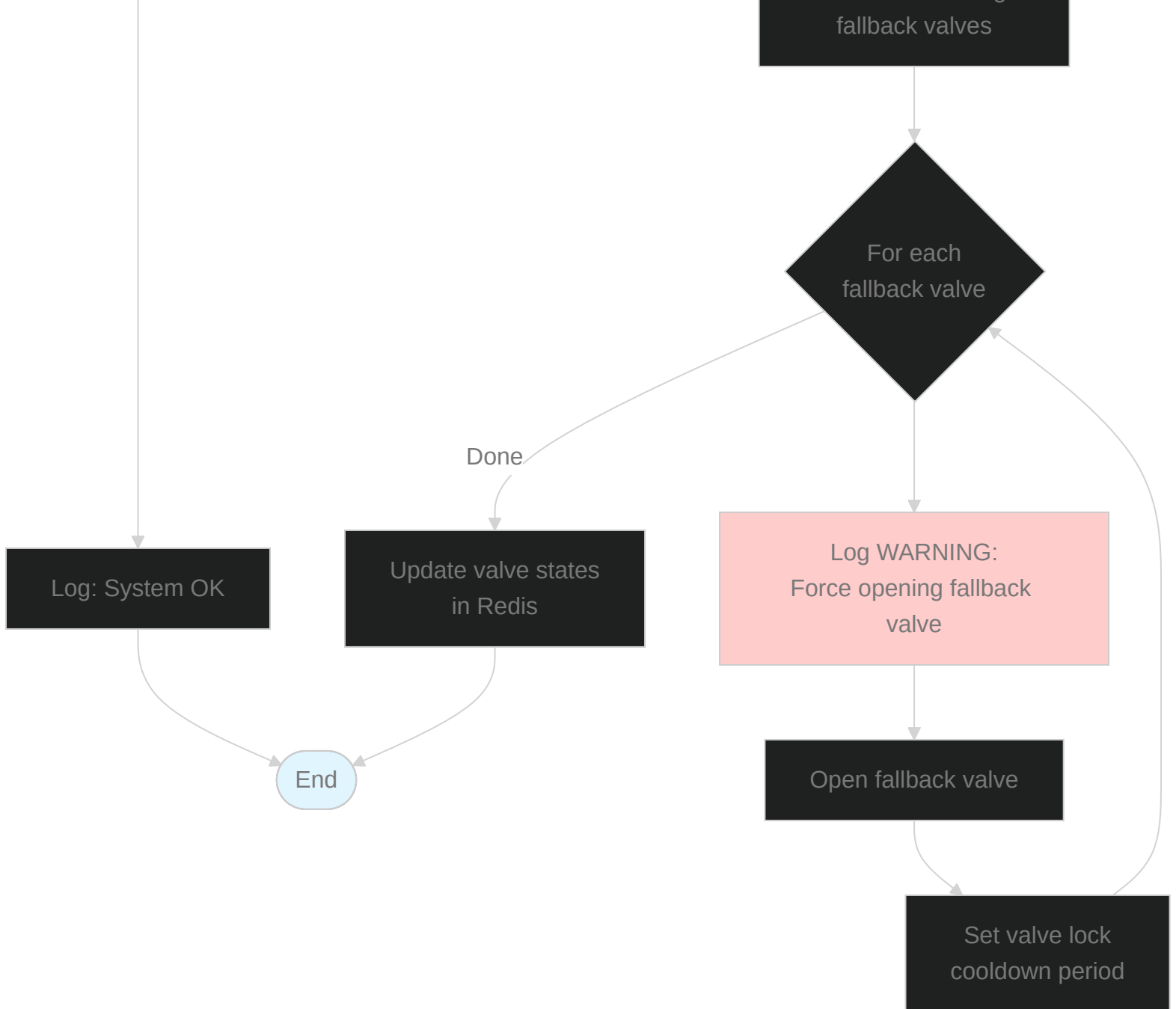


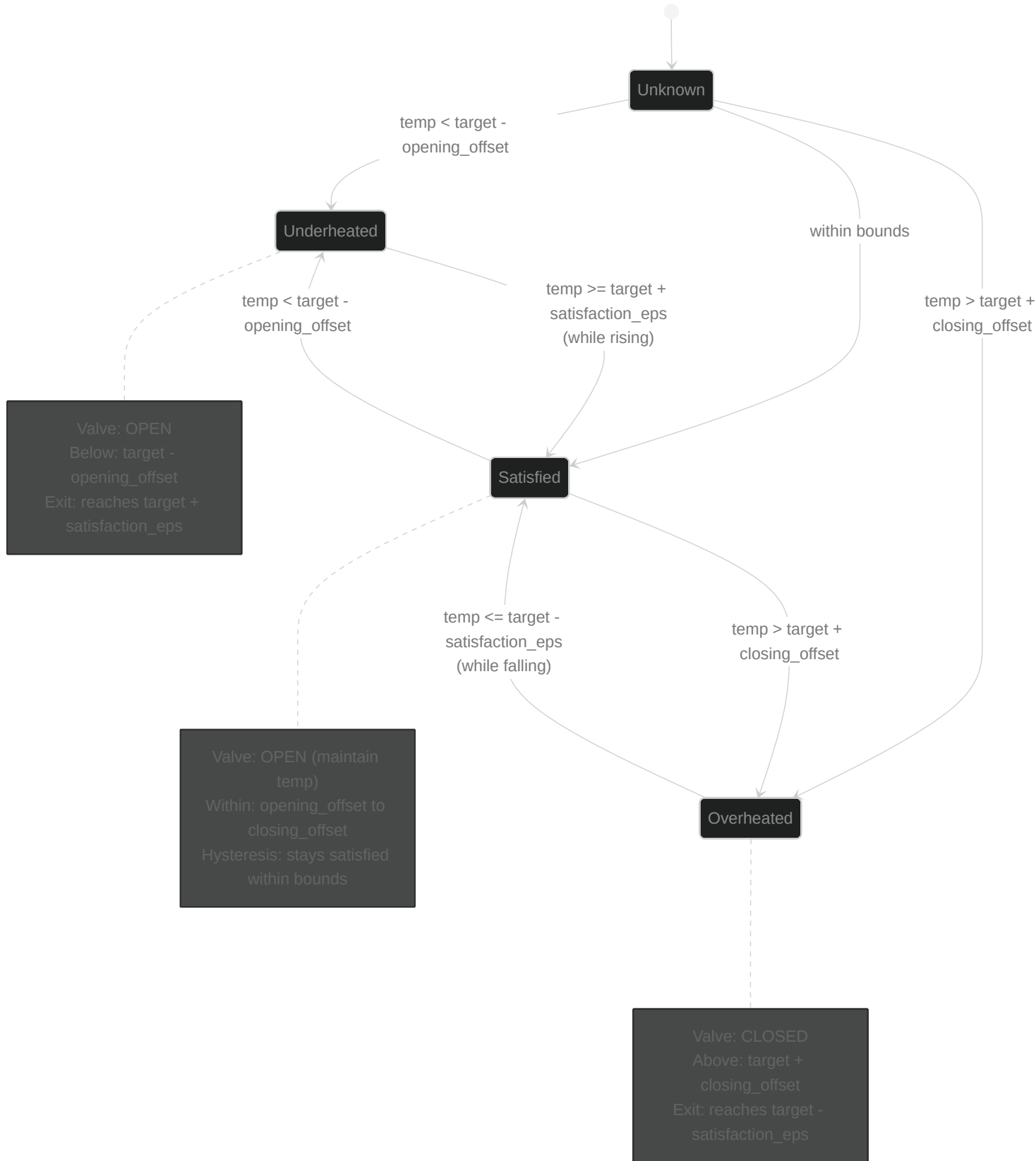


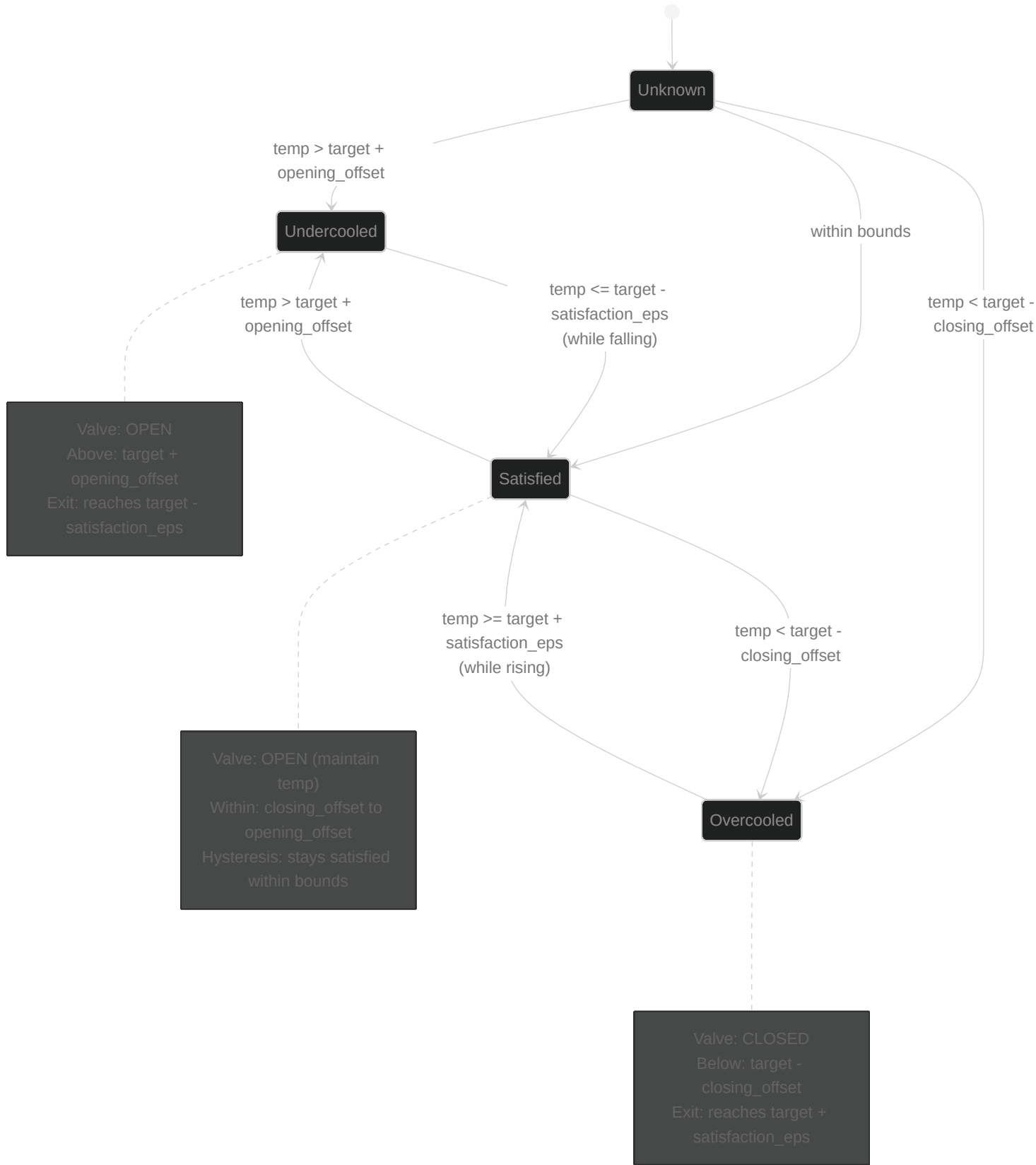












Heating Mode: Target =
opening_offset = 0.3,
closing_offset = 0.3,
satisfaction_eps = 0.1

Target
21.0°C

Satisfied Entry
21.1°C
target + eps

Lower Bound
20.7°C

Underheated Zone

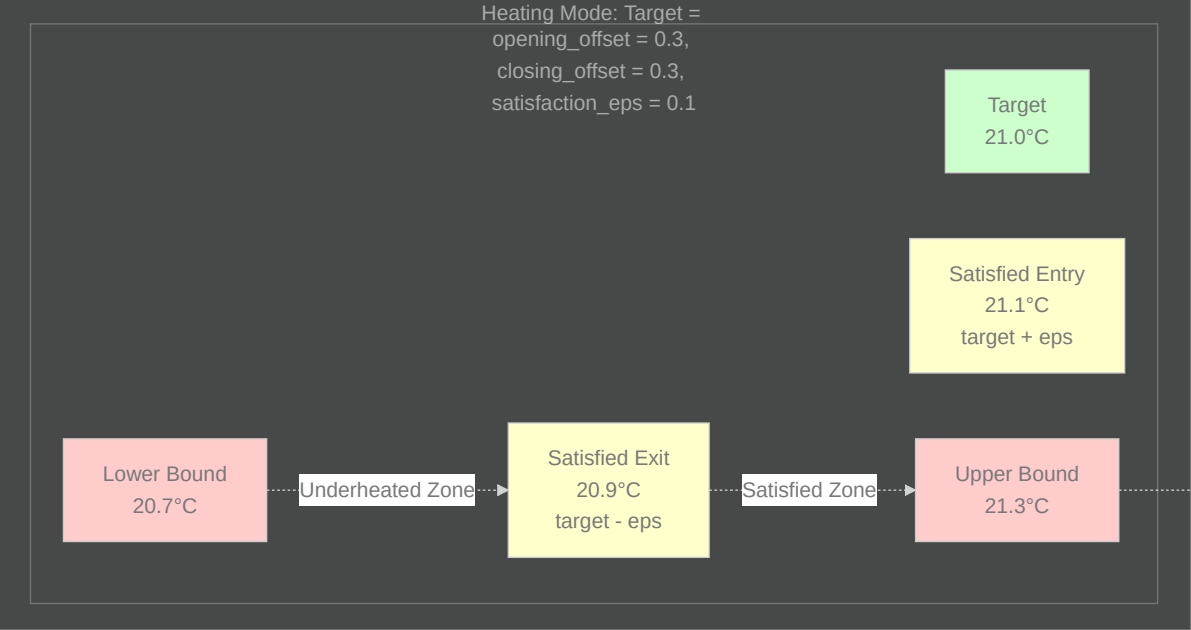
Satisfied Exit
20.9°C
target - eps

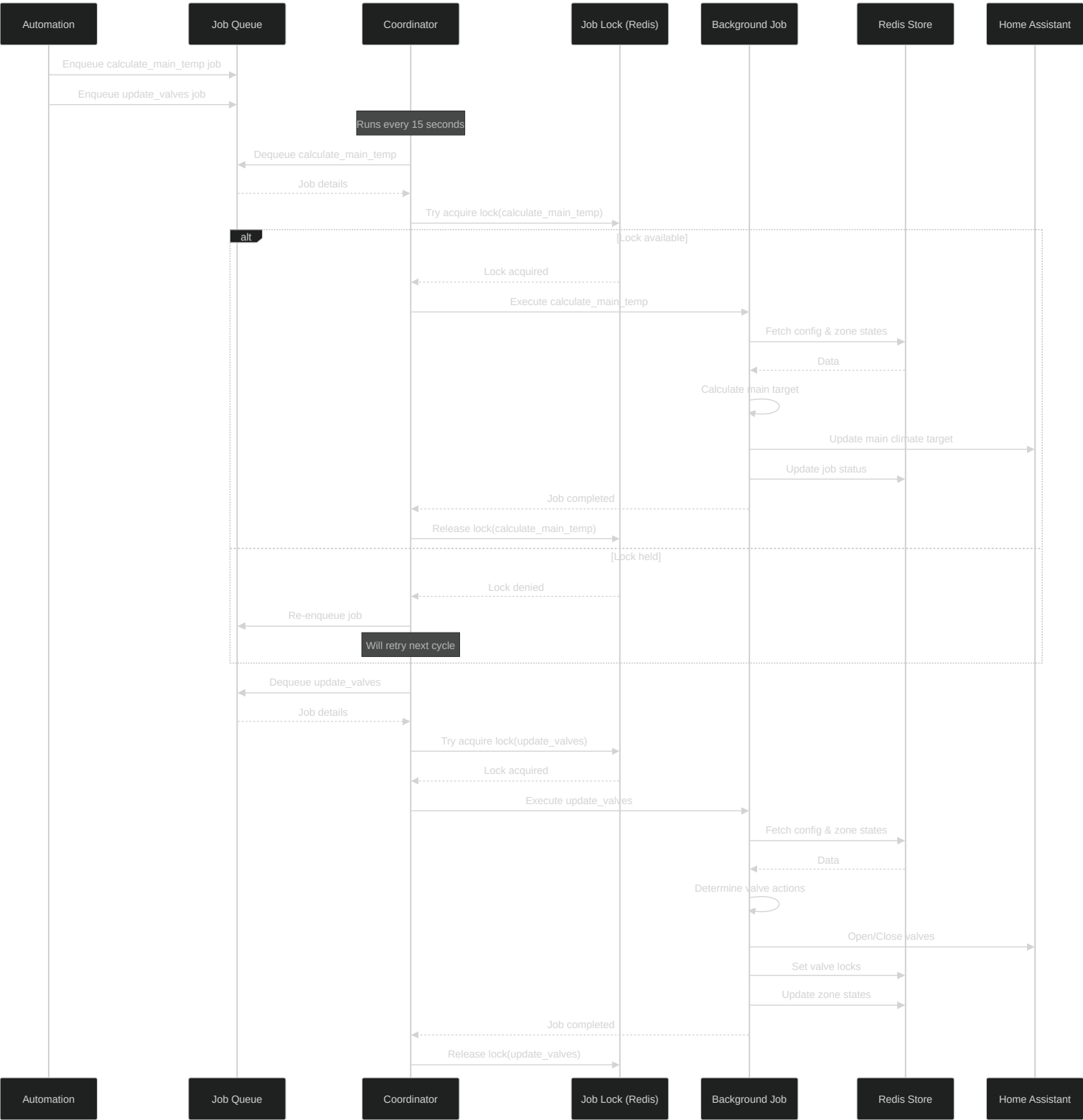
Satisfied Zone

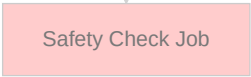
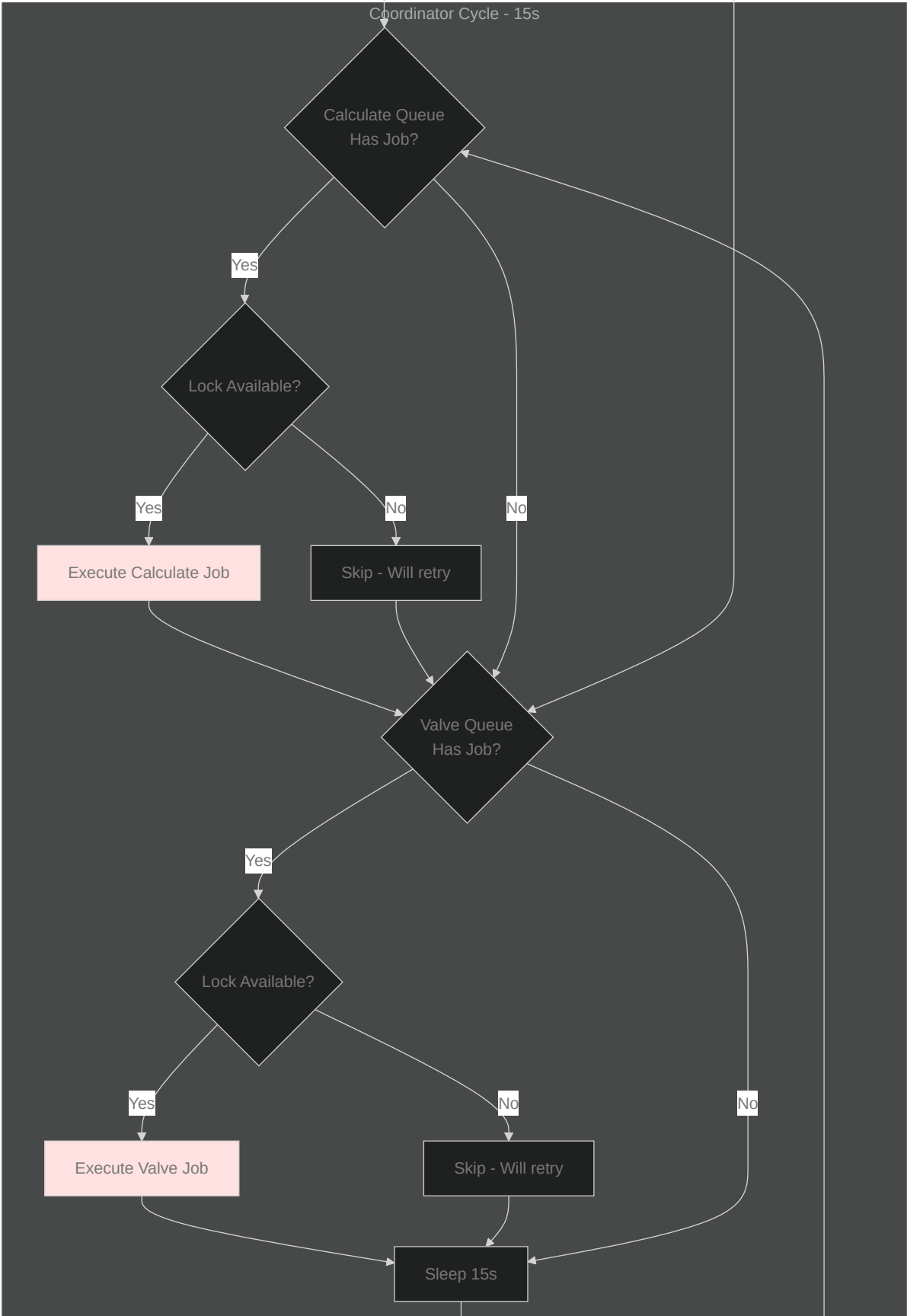
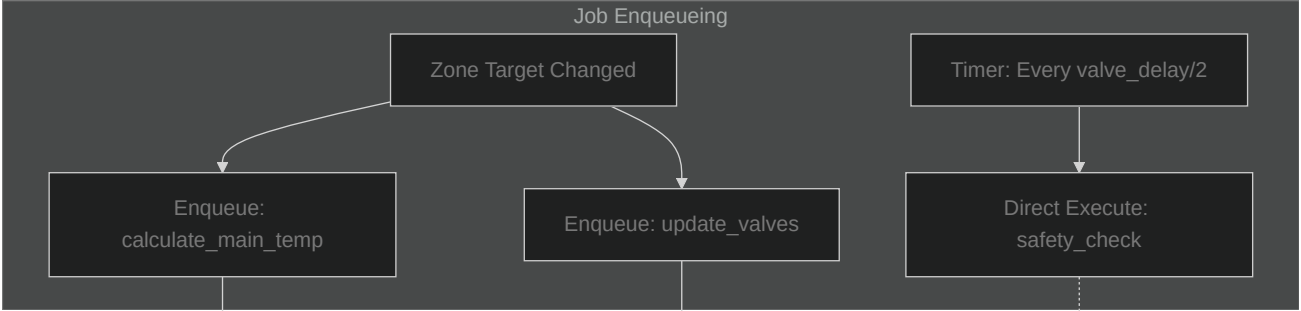
Upper Bound
21.3°C

Overheated Zone

>21.3°C

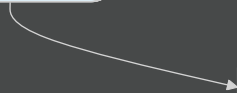






Coordinator Wakes

Update Sensor States
from Redis



CONFIG	
string	main_climate_entity_id
float	main_target_all_zones_satisfied
boolean	use_average_mode
int	min_valves_open
float	main_min_temp
float	main_max_temp
float	main_change_threshold
int	valve_actuation_delay
int	coordinator_interval
float	satisfaction_eps

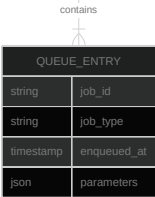
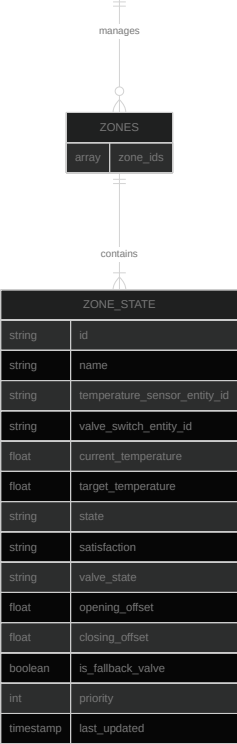
MAIN_CLIMATE	
string	entity_id
float	current_temperature
float	target_temperature
float	outdoor_temperature
string	hvac_mode
string	hvac_action
boolean	multizone_enabled
timestamp	last_updated

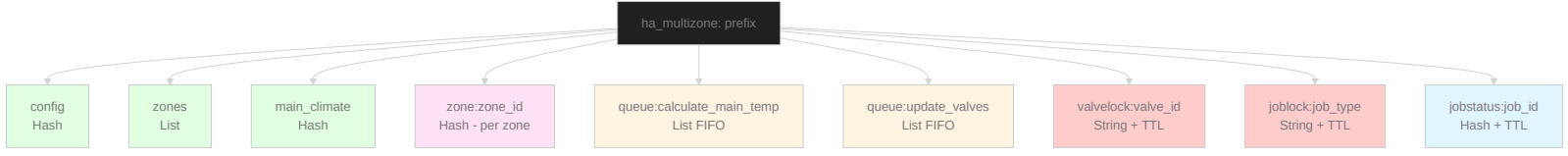
JOB_QUEUES	
string	queue_calculate_main_temp
string	queue_update_valves

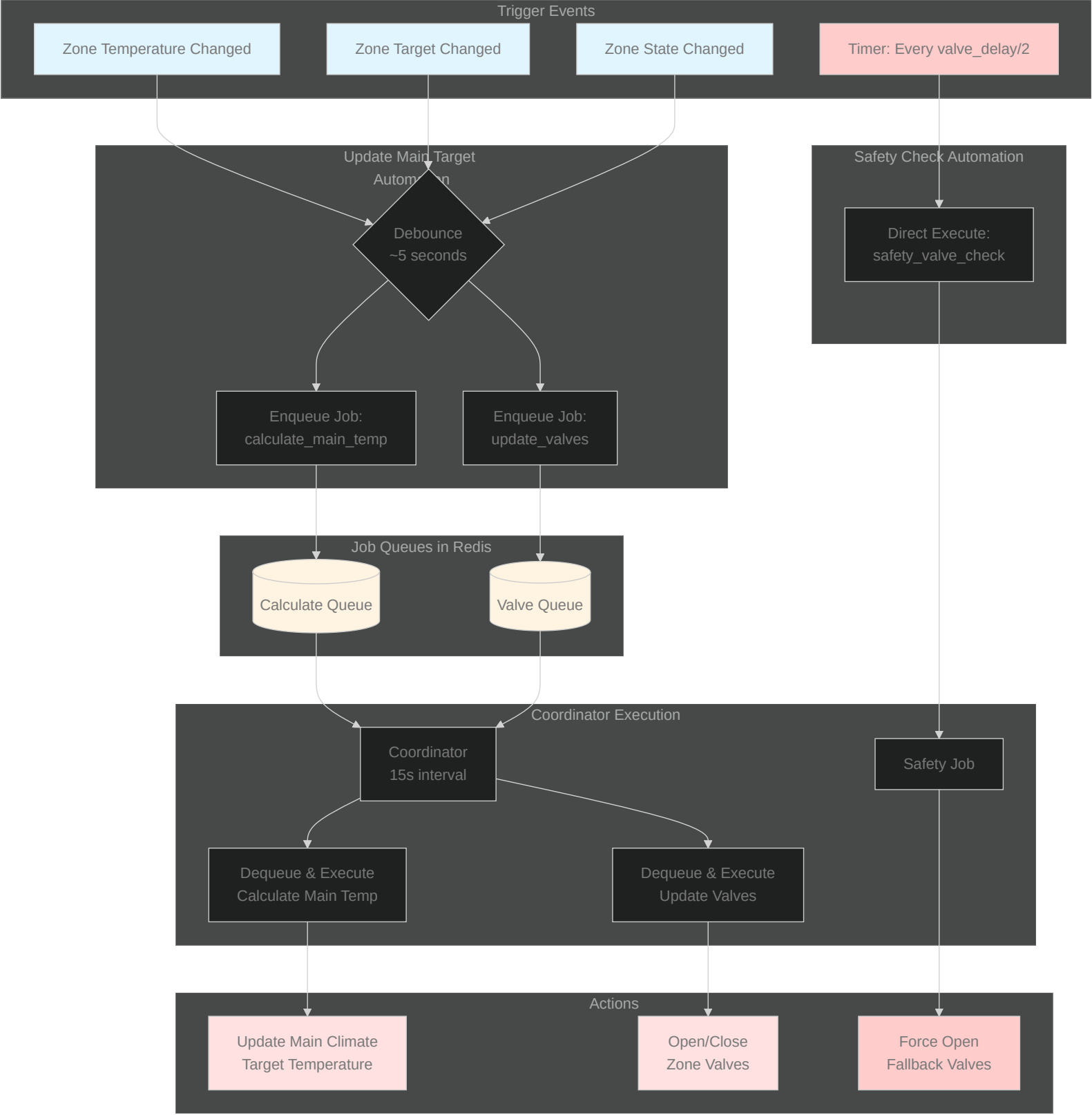
VALVE_LOCKS	
string	valve_id
timestamp	locked_until
string	reason

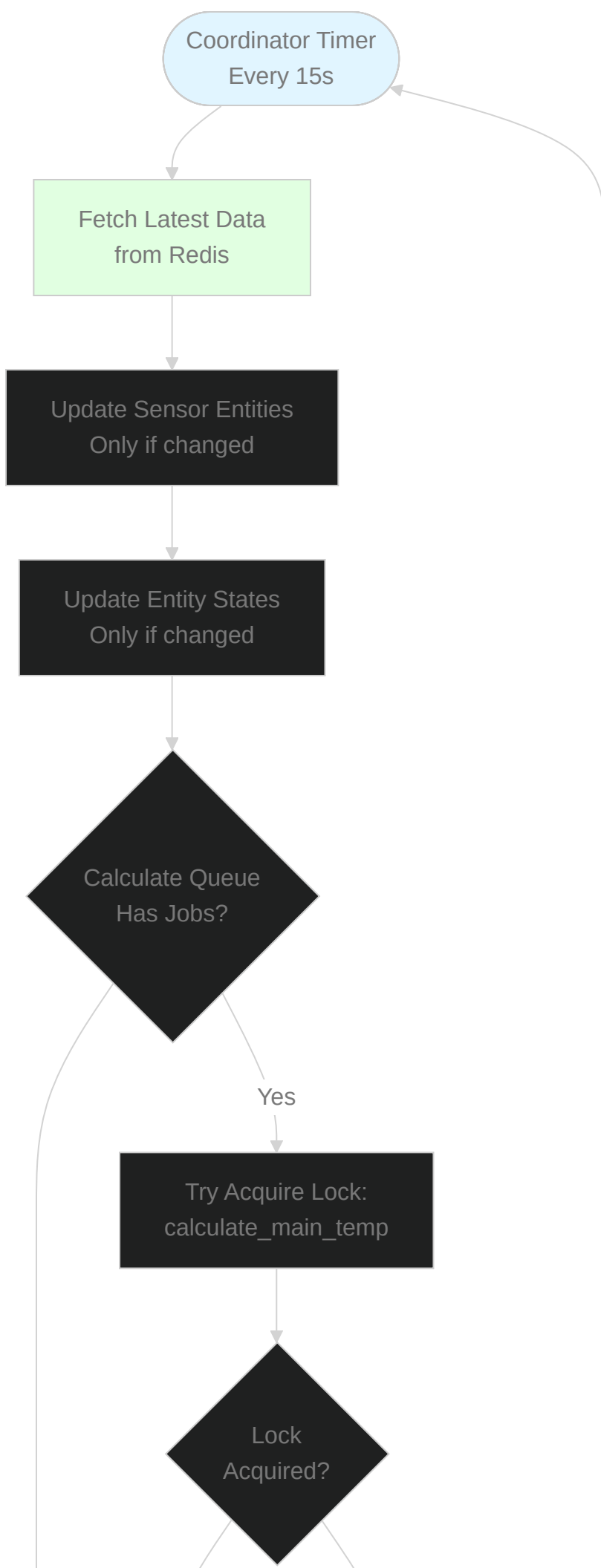
JOB_LOCKS	
string	job_type
timestamp	acquired_at
string	acquired_by

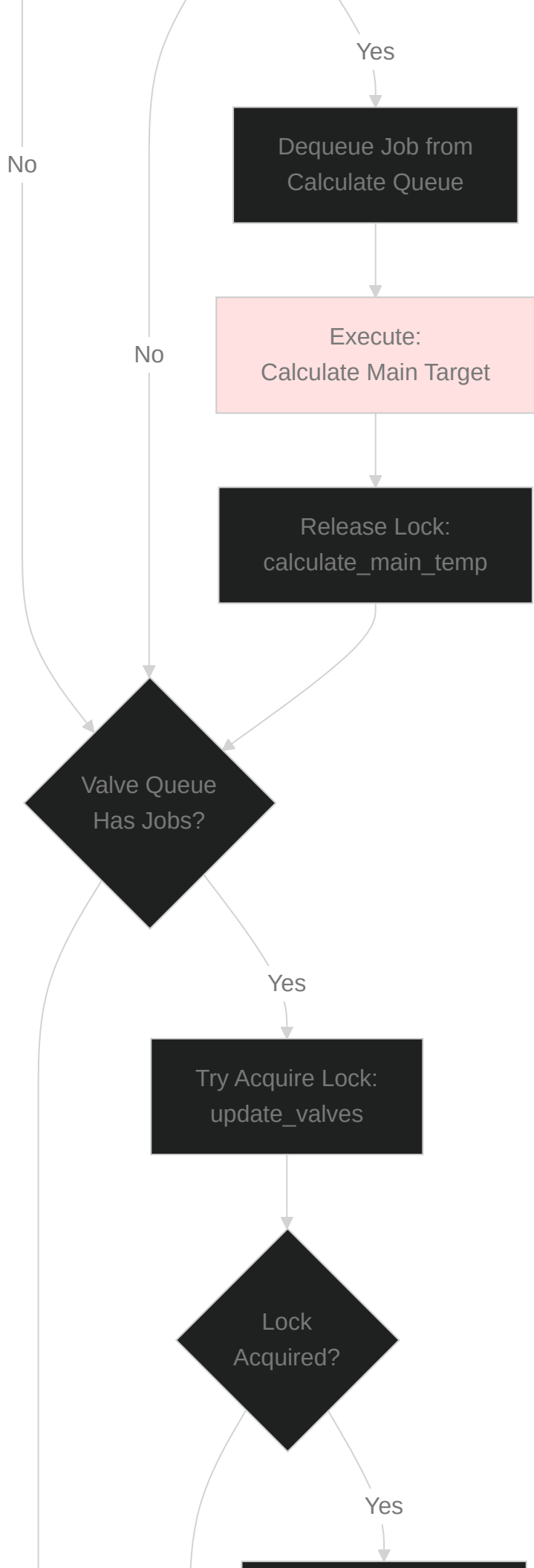
JOB_STATUS	
string	job_id
string	job_type
string	status
timestamp	started_at
timestamp	completed_at
int	duration_ms
int	actions_taken
json	result

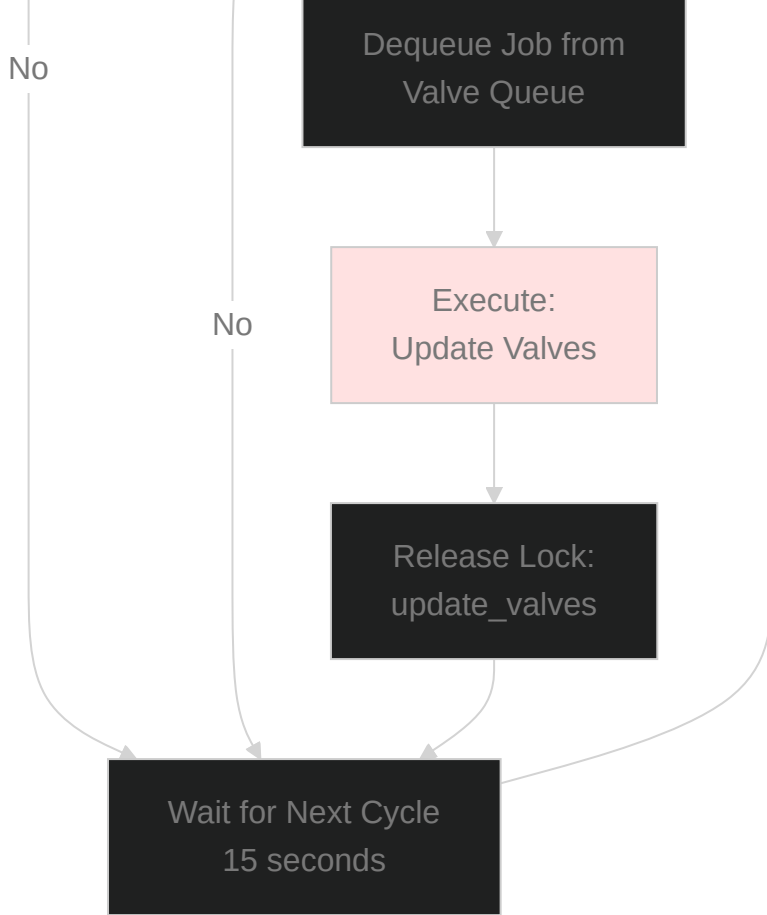


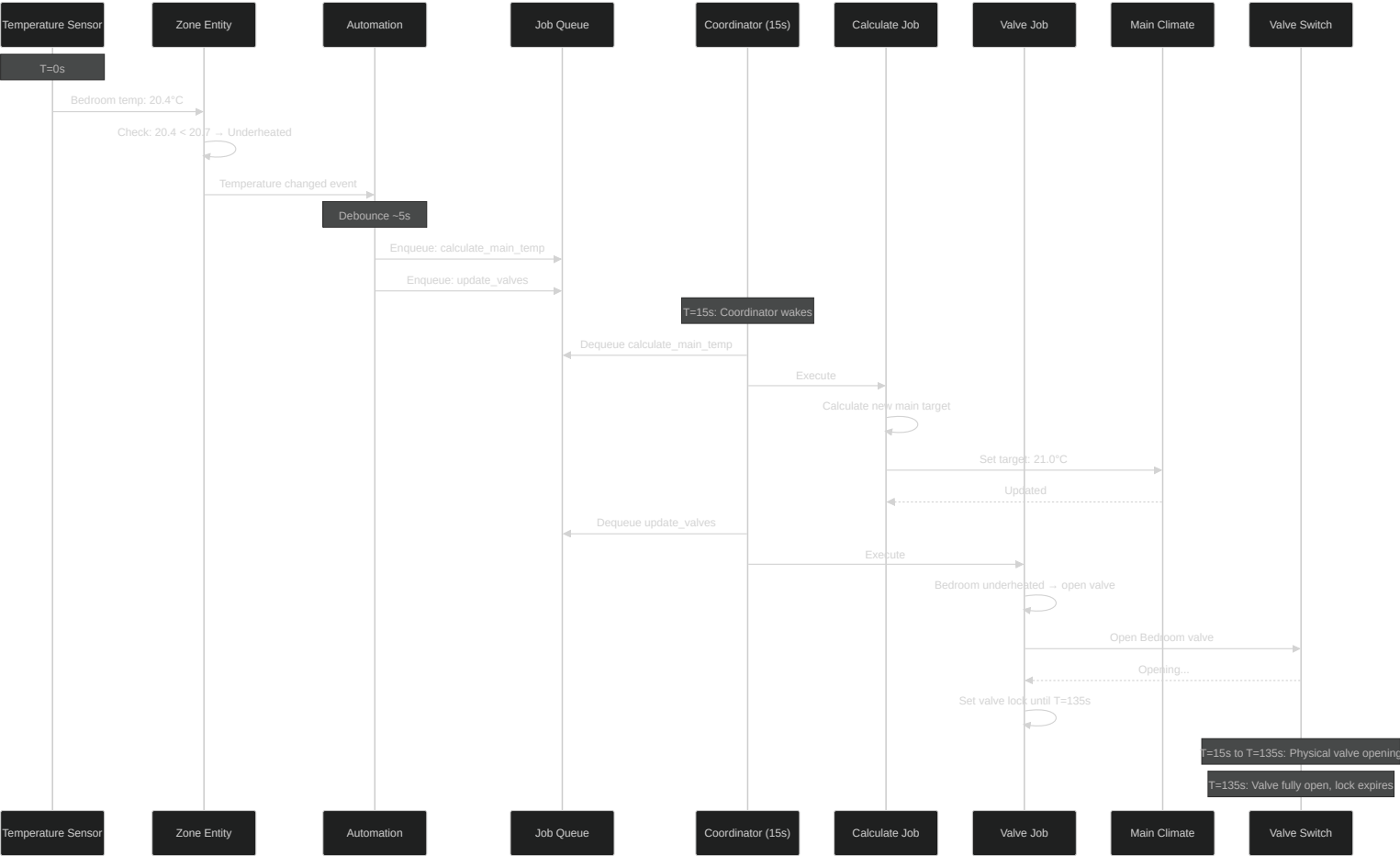














T=60s: All jobs processed

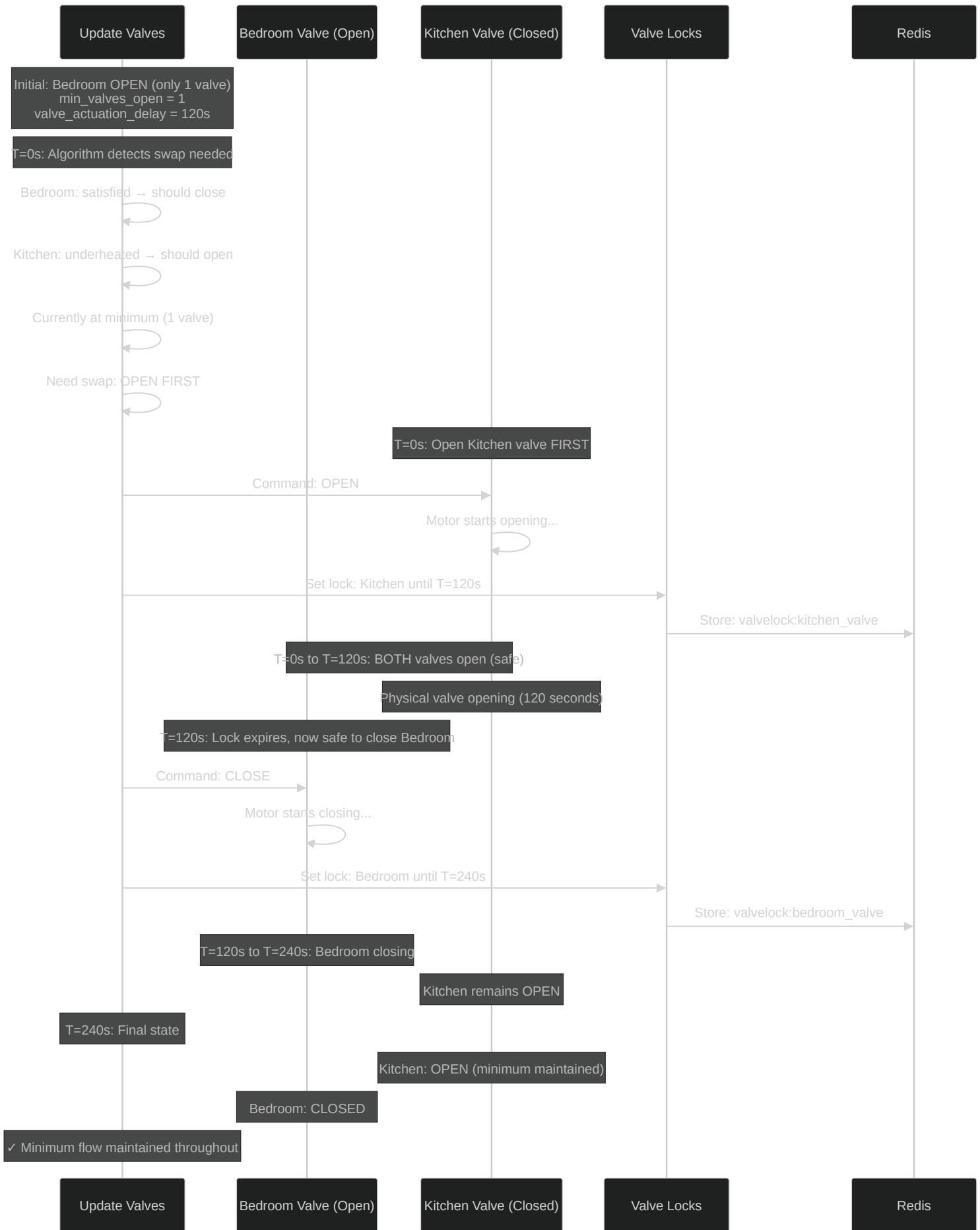
Bedroom Zone

Kitchen Zone

Living Zone

Job Queue

Coordinator



Time: T=0s

Valves Open: 1
Bedroom: OPEN
Kitchen: CLOSED

Time: T=0s Action

OPEN Kitchen valve
Set lock: 120s

Time: T=0-120s

Valves Open: 2
Bedroom: OPEN
Kitchen: OPENING
✓ SAFE: Above minimum

Time: T=120s Action

CLOSE Bedroom valve
Set lock: 120s

Time: T=120-240s

Valves Open: 1+
Bedroom: CLOSING
Kitchen: OPEN
✓ SAFE: At minimum

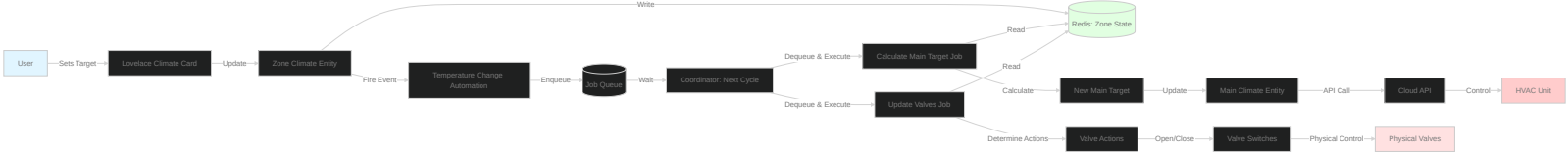
Time: T=240s Final

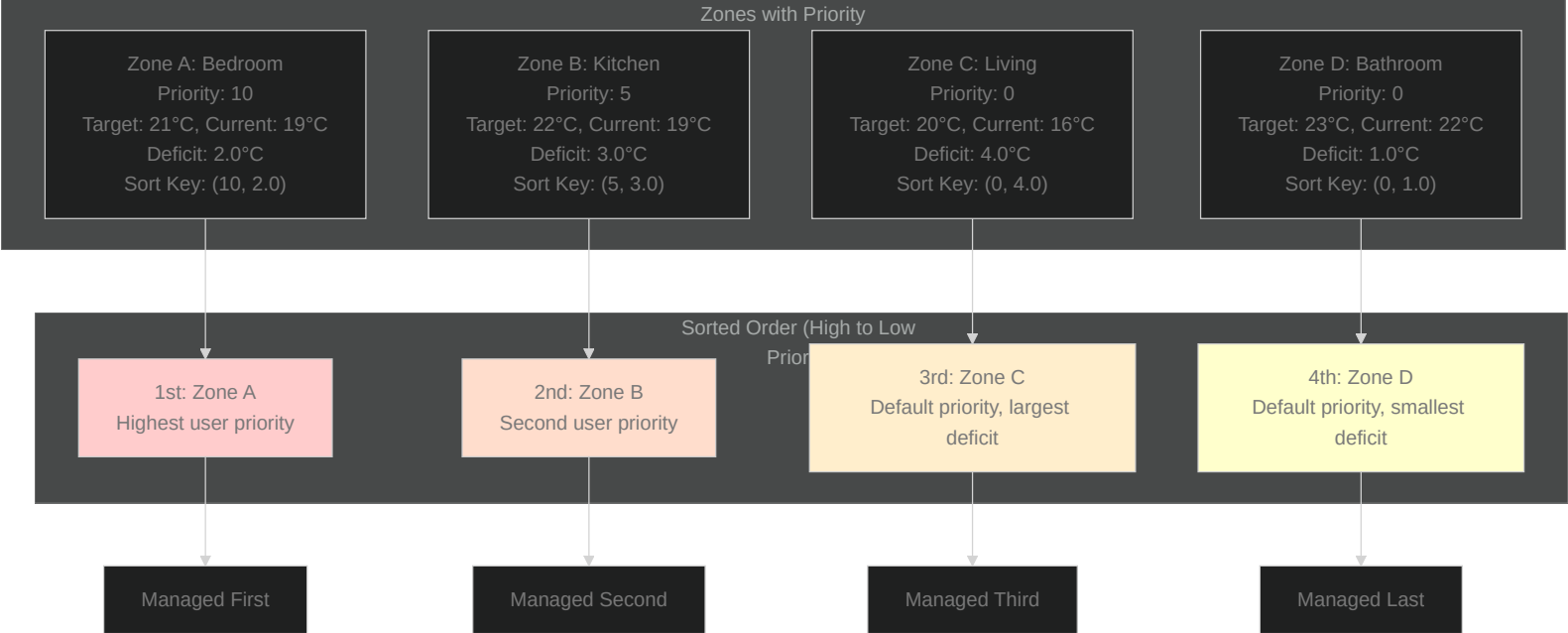
Valves Open: 1

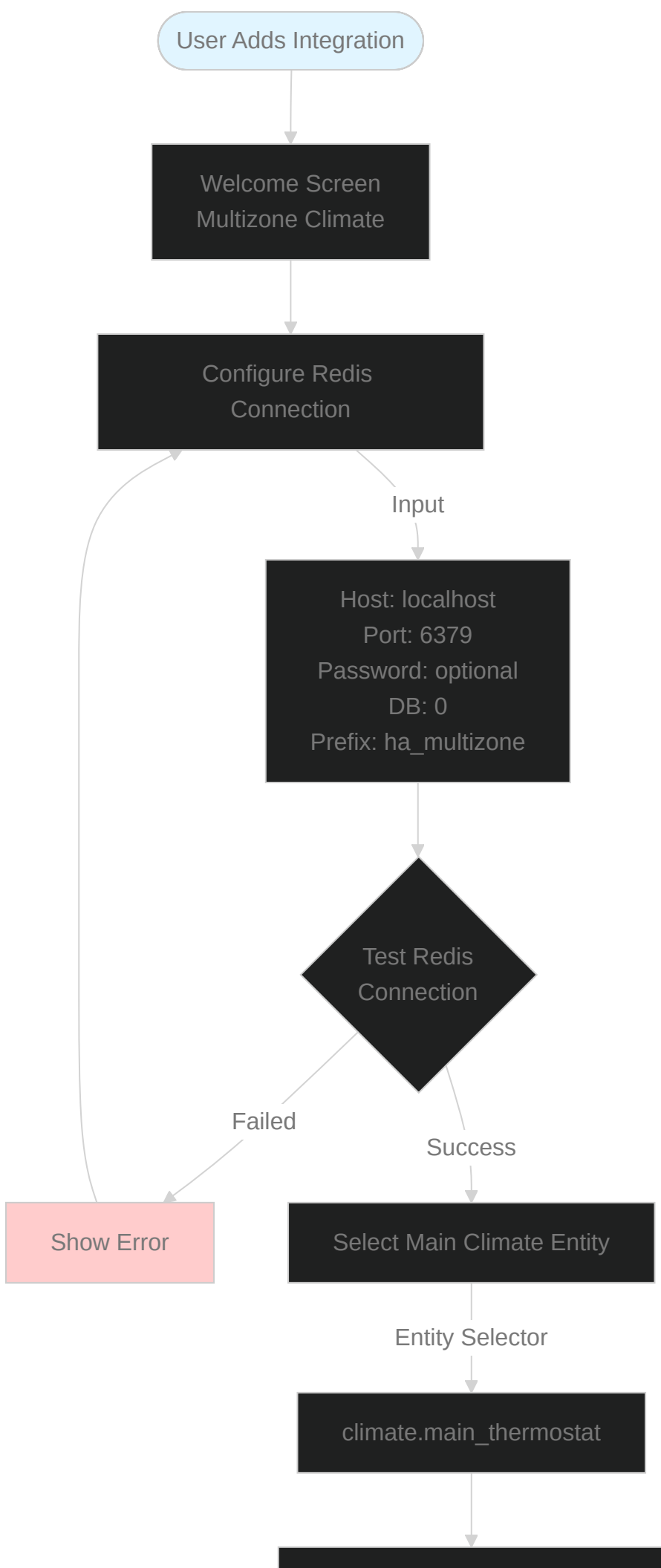
Bedroom: CLOSED

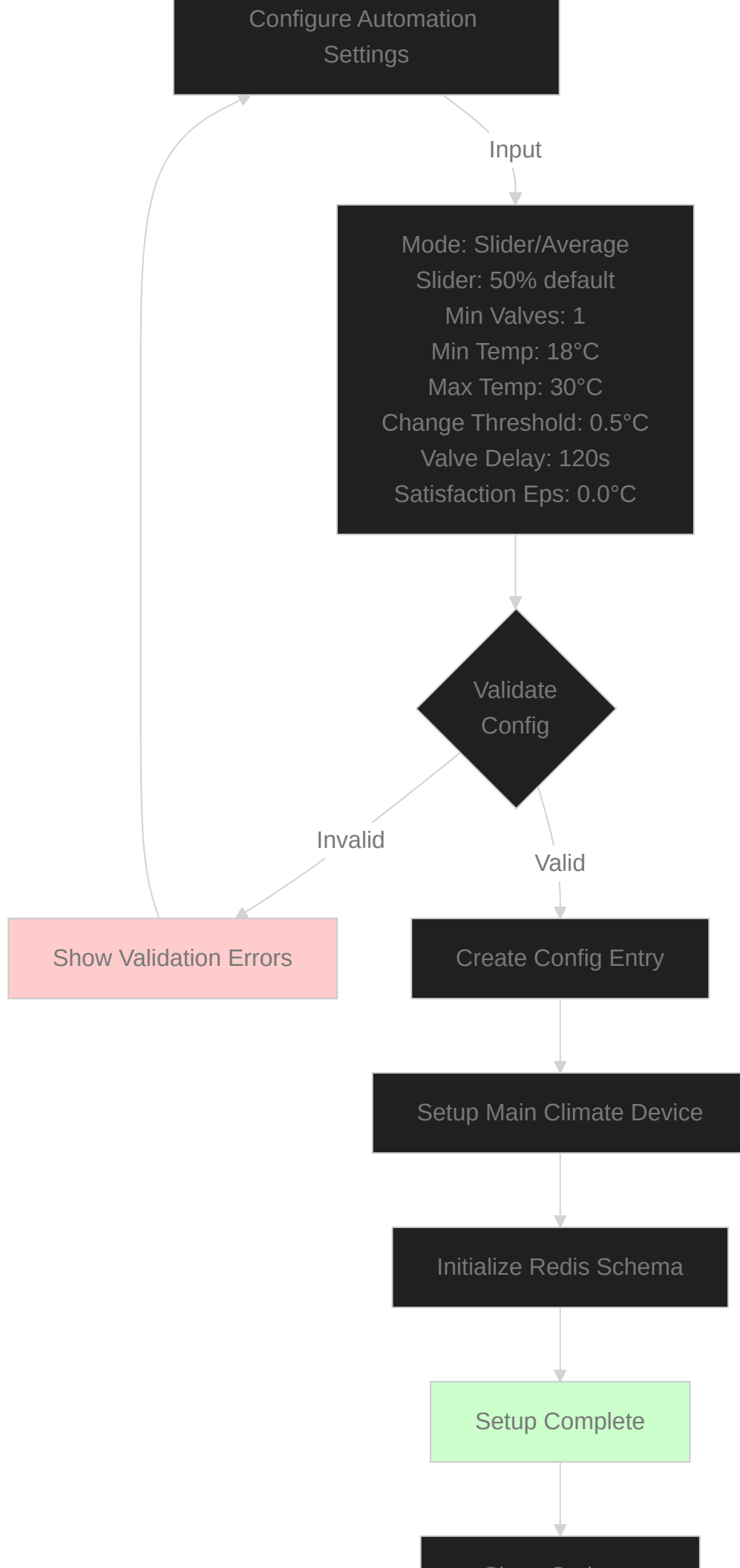
Kitchen: OPEN

✓ SAFE: At minimum









Show Options:
Add Climate Zone

User Adds Zone

Zone Configuration Form

Input

Name: Bedroom
Temp Sensor:
sensor.bedroom_temp
Valve Switch:
switch.bedroom_valve
Target Threshold: 0.1°C
Opening Offset: 0.3°C
Closing Offset: 0.3°C
Is Fallback: false
Priority: 0

Validate
Entities Exist?

No

Yes

Check Fallback
Requirements

Need more fallbacks

OK

Show Error:
Entity not found

Warn: Configure as
fallback?

Create Zone Device

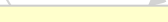
Store Zone in Redis

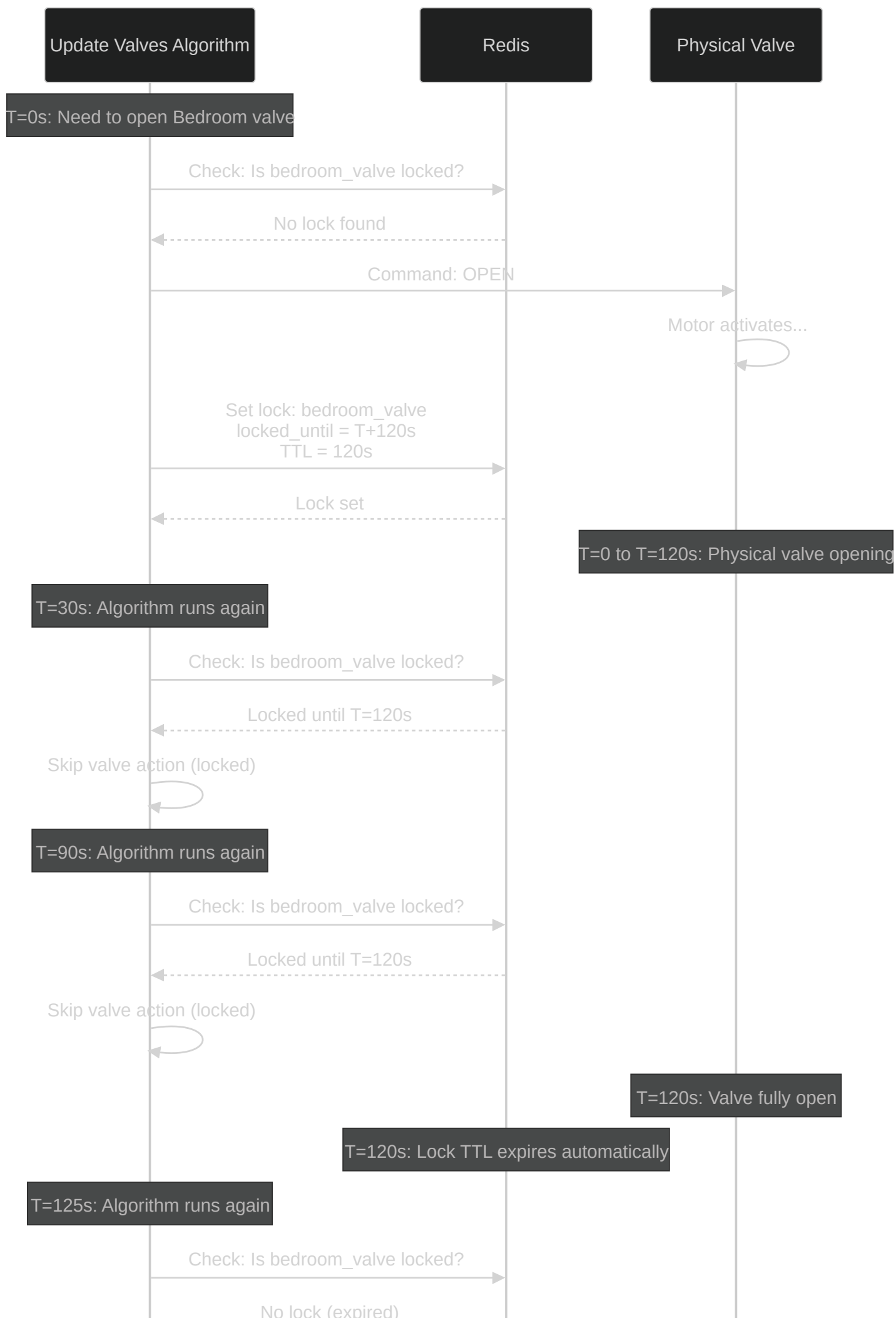
Create Climate Entity

Set Initial State: OFF

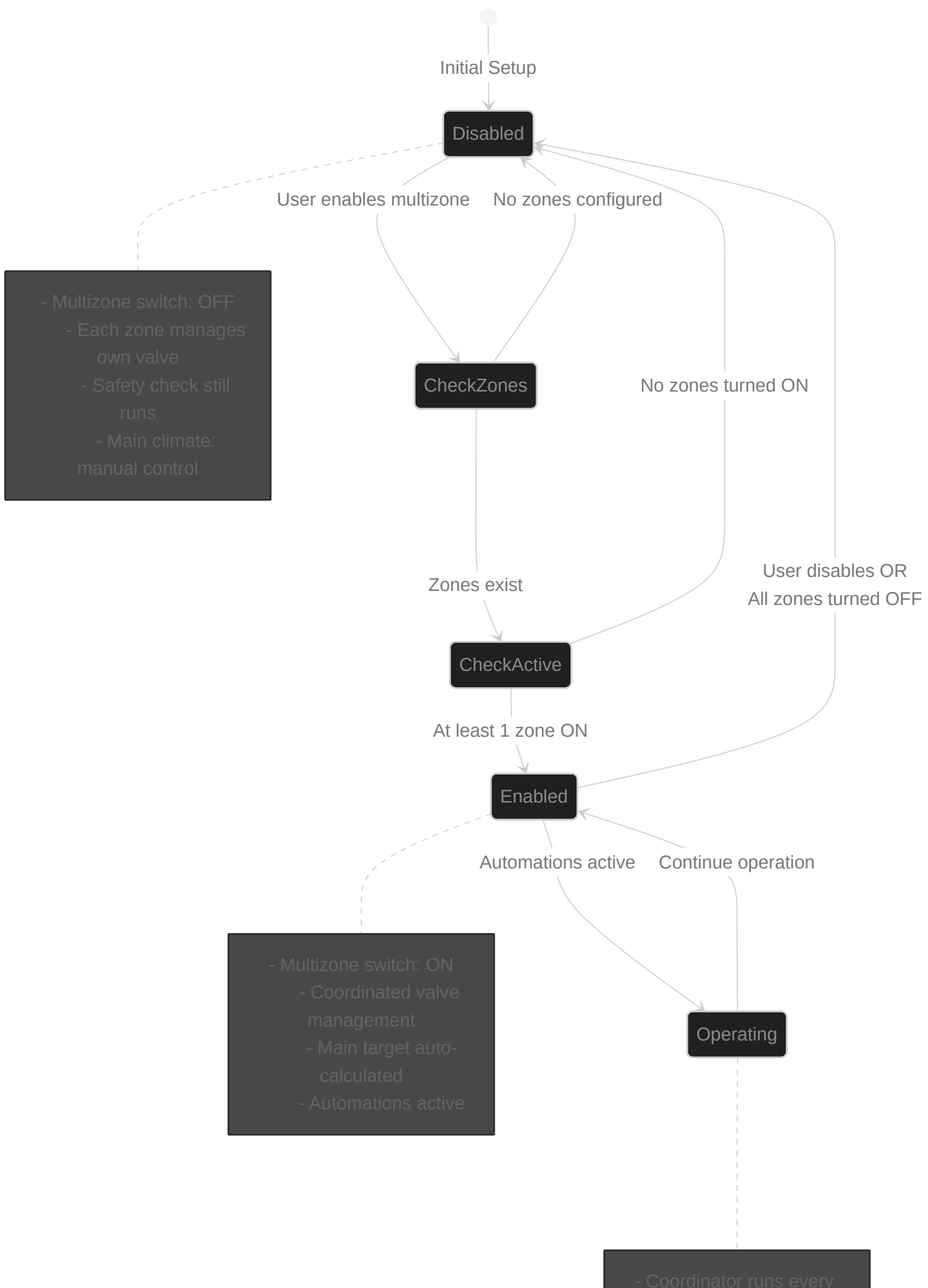
Zone Added

User can now:
Turn zone ON
Set target temp
Enable multizone









15s

- Jobs process from queues
- Valves coordinated
- Main target updated



